

APPLICATION SPECIFICATION

OralCare / Oral Cancer CareConnect

Application Specification

BUILD	pilot, commit <code>c1bbb75</code>
PLATFORMS	Android (full), Web (full except camera capture)
LIVE WEB BUILD	https://oralcare.web.app (the original <code>oral-cancer-pilot-1027-dc3a3.web.app</code> also stays current — both sites receive every deploy)
BACKEND	Firebase Authentication + Cloud Firestore (free Spark plan; no Cloud Storage, no Cloud Functions)
VERIFICATION AT TIME OF WRITING	<code>flutter analyze</code> clean · 174 tests pass · web and debug APK build

Generated 2026-09-22 from `SPECIFICATION.md` . Describes the application as built, not as proposed.
Section 14 carries the clause-by-clause traceability matrices against the source requirement documents.

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How to use this document

This is a reverse-specification: it describes what the application actually does, written so it can be laid side by side with the four source documents.

Sections 1–13 describe the app. **Section 14 is the traceability matrix** — one table per source document, clause by clause, with an implementation status and a pointer into this document. If your goal is "did they build what I asked for", start at section 14 and follow the links back.

Status vocabulary used throughout:

Status	Meaning
Built	Implemented and exercised by a test or a build
Built, limited	Implemented, but with a stated constraint the reader must know about
Recorded only	The data is captured and stored but does not drive any behaviour
Not built	Absent, with the reason given

Source documents referenced:

Short name	File	Role
IPO	Oral_Cancer_Mobile_App_Patient_Doctor_Login.docx	The core Input–Process–Output specification. Sections 1–7, Tables 1–11. Primary contract.
Clinician	Copy 3.docx	Clinician module scope, sections 1–8
CareConnect	Final cHeck.docx	"Oral Cancer CareConnect" feature list, sections 1–12 / modules D–J
Questionnaire	sree cancer questionnaire final (1).docx	Bilingual English–Kannada illustrated questionnaire; source of all Kannada clinical terms

1. System architecture

The IPO document defines four layers (IPO §1). All four exist as distinct code regions:

IPO layer	Implementation	Location
PATIENT LOGIN	20 patient-facing screens	lib/features/patient/ , lib/features/auth/ , plus 6 support modules
RISK & SAFETY ENGINE	Pure function, no I/O, no Flutter dependency	lib/domain/risk_engine.dart
DOCTOR LOGIN	5 clinician screens	lib/features/doctor/
OUTCOME / VALIDATION DATABASE	Aggregation service + read-only screen	lib/domain/validation_service.dart , lib/features/doctor/validation_screen.dart

Layering rule enforced by directory: domain/ holds pure logic and content catalogues with no Firebase or Flutter imports; data/ holds models and repositories; features/ holds screens; state/ holds in-flight controllers. The risk engine is deliberately isolated so it can be re-run over stored answers during validation without touching the network.

Code size: ~20,400 lines of Dart across lib/ and test/ .

2. Roles and access model

Three roles. Two are user-facing; the third is analytical.

Role	How it is granted	What it can do
Patient	Self-registration	Own records only
Doctor	Firebase custom claim <code>role: doctor</code> , set server-side by <code>tools/grant_doctor.mjs</code> — or the pilot enrolment code path (see below)	Read records addressed to them; write clinical assessment and outcome on those records
Coordinator	Firebase custom claim <code>role: coordinator</code> only. Cannot be self-assigned — there is deliberately no profile-field fallback for this claim	Read every shared record in the cohort for validation statistics. Can never write a clinical finding, and is not offered to patients as a sharing recipient

Guest mode (CareConnect §1, "guest-information mode") is a fourth, unauthenticated state: the full assessment flow runs, but the profile cannot be saved and records are session-only. Reminders use the patient ID `GUEST` .

2.1 The enrolment-code weakness — stated plainly

A clinician can self-register by typing the code `ORAL-PILOT-2026` , and the Firestore rules contain a fallback that trusts `role: doctor` written on a user's own profile document. **Both are security holes:** the code ships inside the app bundle and can be read out of it, and a patient can write `role: doctor` onto their own profile. They were restored at the project owner's explicit request so the pilot can onboard clinicians without a service-account key.

Mitigations in place: - Build with `--dart-define=ALLOW_ENROLMENT_CODE=false` to remove the path entirely — the Doctor login stops offering account creation and the repository refuses it. Covered by tests that run under either build. - Override the code without a source edit: `--dart-define=ENROLMENT_CODE=...` - The coordinator claim has no such fallback, so cohort-wide statistics remain claim-only.

Before real-world deployment, the enrolment path and the `hasDoctorProfile()` rule fallback must both be removed.

3. Complete screen inventory

26 screens. Patient side first, then clinician.

#	Screen	Class	File
Entry and account			
1	Role selection	RoleSelectScreen	features/auth/role_select_screen.dart
2	Login	LoginScreen	features/auth/login_screen.dart
3	Registration	RegisterScreen	features/auth/register_screen.dart
4	Consent	ConsentScreen	features/patient/consent_screen.dart
Patient hub			
5	Check Hub (tab 1)	PatientHomeScreen	features/patient/patient_home_screen.dart
6	My profile	ProfileScreen	features/patient/profile_screen.dart
7	Reminders	RemindersScreen	features/patient/reminders_screen.dart
Assessment flow			
8	Step 1 — Risk factors	RiskAssessmentScreen	features/patient/risk_assessment_screen.dart
9	Safety check (conditional)	RedFlagScreen	features/patient/red_flag_screen.dart
10	Step 2 — Self-examination	SelfExamScreen	features/patient/self_exam_screen.dart
11	Step 3 — Record the finding (conditional)	LesionScreen	features/patient/lesion_screen.dart
12	Your result	ResultScreen	features/patient/result_screen.dart
13	Past assessment detail	AssessmentDetailScreen	features/patient/assessment_detail_screen.dart
14	Clinical reference photos	LesionReferenceDialog	features/patient/lesion_reference_dialog.dart
CareConnect support modules			
15	Education (tab 2)	EducationScreen	features/education/education_screen.dart
16	DigiLocker (tab 3)	DigiLockerScreen	features/records/digilocker_screen.dart
17	Cessation (tab 4)	CessationScreen	features/cessation/cessation_screen.dart
18	Care & Rehab hub (tab 5)	_CareAndRehabHub	features/patient/patient_home_screen.dart
19	Screening centres & referrals	ReferralScreen	features/referral/referral_screen.dart
20	Post-treatment rehabilitation	RehabilitationScreen	features/rehabilitation/rehabilitation_screen.dart
21	Emergency guidance	EmergencyScreen	features/emergency/emergency_screen.dart
Clinician			
22	Patient queue	DoctorHomeScreen	features/doctor/doctor_home_screen.dart
23	Patient record review	PatientRecordScreen	features/doctor/patient_record_screen.dart

#	Screen	Class	File
24	Clinical assessment entry	ClinicalAssessmentScreen	features/doctor/clinical_assessment_screen.dart
25	Clinical outcome entry	OutcomeScreen	features/doctor/outcome_screen.dart
26	Algorithm validation	ValidationScreen	features/doctor/validation_screen.dart

3.1 Bottom navigation

Five tabs: Check Hub · Education · DigiLocker · Cessation · Care & Rehab .

3.2 Routing gate

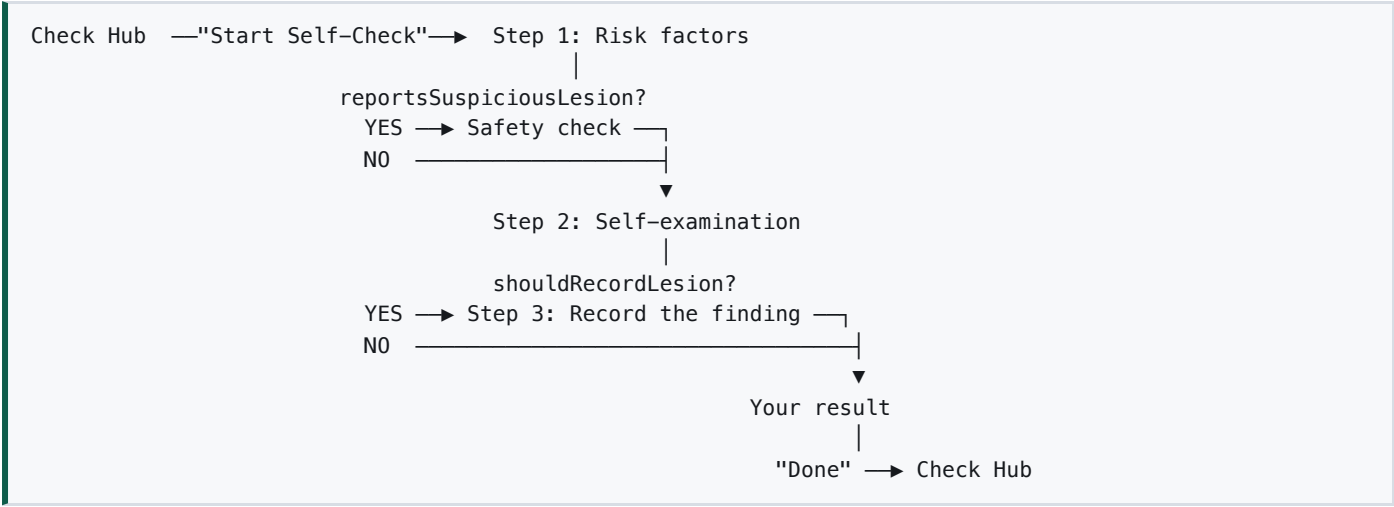
_RootRouter in lib/app.dart decides in this order:

- 1. Session initialising → splash
- 2. No user → screen 1 (role selection)
- 3. role == doctor → screen 22 (queue)
- 4. !consent.appAndSelfExam → screen 4 (consent)
- 5. Otherwise → screen 5 (Check Hub)

Step 4 is the enforcement point for IPO Table 1 "Proceed only when consent = Yes". **There is no code path that reaches the risk assessment without it** — the gate is in the router, not in a button handler, so it cannot be bypassed by navigation.

4. Patient assessment flow

4.1 Navigation order and branch rules



Branch predicates (`lib/state/assessment_flow.dart`):

- `reportsSuspiciousLesion` = answer to `suspicious_lesion` is `yes`
- `shouldRecordLesion` = `anySiteAbnormal` || `reportsSuspiciousLesion`

A consequence worth knowing: a patient who reports a suspicious lesion on Step 1 always reaches Step 3, even if they mark no abnormal site on Step 2. This is intentional — they already said something is wrong, so the app must let them describe it.

The result screen has no back button. Answering "No" to `suspicious_lesion` after previously answering "Yes" erases the duration answer **and every** **ticked red flag**, so a stale flag can never reach the engine.

4.2 Registration fields (IPO Table 1 / CareConnect §1)

Field	Control	Required	Validation
Username	text	Yes	≥3 chars, <code>^[A-Za-z0-9._-]+\$</code>
Password	text, obscured	Yes	≥8 chars, ≥1 letter, ≥1 digit
Confirm password	text, obscured	Yes	must match
Name	text	No	—
Patient ID	text	No	<code>^[A-Za-z0-9/_-]{3,32}\$</code> ; auto-generated if blank
Age in years	numeric	Yes	0–120
Gender	dropdown	Yes	Male / Female / Transgender / Prefer not to say
Mobile number	phone	No	≥10 digits if entered
Town or city	text	No	—
Relevant medical history	text, 2 lines	No	—
Emergency contact name	text	No	—
Emergency contact number	phone	No	≥10 digits if entered

Age and gender are hard-required: age is a scored risk variable, so an absent value would silently score 0. The field is labelled **Gender**, not "Sex" — the IPO document says "Sex, where used", CareConnect §1 says "gender", and the Questionnaire says `GENDER` / `Gender`. Gender wins. Accounts created before the rename are still read correctly: `AppUser.fromFirestore` falls

back to the legacy `sex` key.

Patient ID format: `0C-{year}-{6 chars}` , e.g. `0C-2026-4F9A2C` , derived from the Firebase uid. Uniqueness is proved by a creation-only reservation document at `patient_ids/{patientId}` rather than by querying other users' profiles — a patient must not be able to read the user collection. A failed profile write rolls back the Auth account so a username is never stranded.

4.3 Consent (IPO Table 1)

Three independent switches, each a hard gate:

Consent	Gate effect	Firestore key
Use the app and examine my own mouth	Required. Nothing downstream runs without it	<code>consent_app</code>
Store a photograph of a finding	Enables/disables photograph capture	<code>consent_photo</code>
Share my record with a doctor	Enables/disables addressing a record to a clinician	<code>consent_share</code>

Reachable again at any time from the Check Hub shield icon, so consent can be withdrawn as easily as given. Withdrawal is immediate and real: clearing photograph consent strips *all three* routes to the image (see §4.8), and clearing share consent removes the record from the clinician's queue on the next load.

4.4 Step 1 — Risk-assessment variables (IPO Table 2)

15 variables. All applicable ones are **required**; Continue is blocked with a count of what is missing. Scores are the provisional pilot weights from IPO Table 2.

Key	Question	Role	Options → score
<code>age_band</code>	Age	scored, derived	<code><40</code> →0 · <code>40-49</code> →1 · <code>50-59</code> →2 · <code>≥60</code> →3
<code>smoking</code>	Smoking	scored	Never→0 · Former→1 · Current occasional→2 · Current regular→3
<code>smokeless_tobacco</code>	Smokeless tobacco	scored	Never→0 · Former→1 · Current→3
<code>areca_betel</code>	Areca / betel nut	scored	Never→0 · Former→1 · Occasional current→2 · Regular current→3
<code>gutkha_pan_masala</code>	Gutkha / pan masala	exposure only	No · Past · Current — all unweighted
<code>exposure_duration</code>	Duration of tobacco / areca exposure	scored, conditional	<code><5 y</code> →0 · <code>5-10 y</code> →1 · <code>>10 y</code> →2
<code>use_frequency</code>	Frequency of use	scored, conditional	Occasional→0 · Once daily→1 · Several times daily→2
<code>alcohol</code>	Alcohol	scored	None→0 · Occasional→1 · Regular→2
<code>previous_opmd</code>	Previous OPMD	scored	No→0 · Yes→ 4 · Don't know→ unknown
<code>previous_oscc</code>	Previous oral cancer (OSCC)	scored	No→0 · Yes→ 5 (+ override)
<code>family_history</code>	Family history of cancer	exposure only	No · Yes unweighted · Don't know unknown
<code>immunosuppression</code>	Weakened immune system	exposure only	No · Yes unweighted · Don't know unknown
<code>suspicious_lesion</code>	Suspicious oral lesion or symptom right now?	red-flag gate	No · Yes
<code>lesion_duration</code>	How long has it been present?	referral trigger, conditional	<code><2 weeks</code> · <code>≥2 weeks</code>
<code>neck_lump</code>	New or persistent lump in the neck	referral trigger	No · Yes

Maximum possible score: 27.

Age is not asked here — it is read from the profile and mapped to a band, then displayed read-only. Duration and frequency appear only when some tobacco or areca exposure is reported; otherwise a note explains their absence. `lesion_duration` appears only when a suspicious lesion is reported.

Two deliberate design choices in this table:

- **"Don't know" is retained as unknown, never scored as zero.** The unknown keys are stored on the record and listed to both the patient and the clinician. Scoring an unknown as absent would quietly bias the score downward and make the record unauditible.
- **Unweighted variables score zero and are labelled as such.** `gutkha_pan_masala` is unweighted because IPO Table 2 says "final weight to be validated". `family_history` and `immunosuppression` come from CareConnect module D, which lists them with no weights at all. Inventing a weight would move patients between bands on a guess. A test enforces that adding these answers does not change the total.

4.5 Safety check — red flags (IPO §2.3)

13 items, all optional switches:

#	Item	Source
1	Non-healing ulcer or sore	IPO §2.3
2	Red patch	IPO §2.3
3	White patch	IPO §2.3
4	Red-and-white patch	IPO §2.3
5	Lump or thickening	IPO §2.3
6	Unexplained bleeding	IPO §2.3
7	Persistent numbness	IPO §2.3
8	Difficulty swallowing	IPO §2.3
9	Restricted tongue or jaw movement	IPO §2.3
10	Persistent neck lump	IPO §2.3
11	Teeth loosening without an obvious cause	CareConnect §3
12	Difficulty speaking	CareConnect §3
13	Unexplained weight loss	CareConnect §3

IPO §2.3 lists ten; items 11–13 close the gap against CareConnect §3. In Kannada mode the localised term is the title and the English term stays visible underneath, because that is the term a clinician will ask about.

4.6 Step 2 — Guided self-examination (IPO Table 5 / CareConnect §3)

8 sites. IPO Table 5 lists seven; `throat` is added from CareConnect §3 ("The back of the throat, where feasible").

Key	Site	Instruction
<code>lips</code>	Lips	Look for a sore, patch or lump on both lips.
<code>inner_cheeks</code>	Inner cheeks	Inspect both cheeks.
<code>gums</code>	Gums	Look for any patch or swelling.
<code>tongue</code>	Tongue	Inspect the top, both sides and underside.

Key	Site	Instruction
floor_of_mouth	Floor of mouth	Inspect under the tongue.
palate	Palate	Inspect the roof of the mouth.
throat	Back of the throat	Open wide and look as far back as you comfortably can. Skip this if you cannot see clearly.
neck	Neck	Feel for any new lump.

Per site: two switches — *"I examined this site"* and, once examined, *"I found something abnormal"*. Marking abnormal forces examined to true. Each site carries an illustration (`assets/images/1.png` – `8.png`) and, when marked abnormal, a button into the clinical photo gallery for comparison.

All eight sites now carry an illustration. `assets/images/8.png` (back of the throat) was added after clinical review; it is an intraoral oropharyngeal view matching the style of the palate image, showing a single midline uvula, symmetric tonsillar pillars, the palatine tonsils and the posterior pharyngeal wall, with healthy mucosa throughout.

The fallback behaviour is retained for any site added in future: `_imageForSite` returns null rather than a default, and the card renders a labelled placeholder — *"No illustration for <site> yet. Follow the written instruction below."* It never substitutes another site's image, because showing the wrong anatomy is worse than showing none.

The illustrations are AI-generated, which removes any stock-photography licensing question but introduces a different one: image models are unreliable on mouth anatomy. Each intraoral image should be checked for a single midline uvula, symmetric tonsillar pillars, plausible tooth count and uniformly healthy mucosa before it ships, because these pictures are the reference a patient compares their own mouth against — a malformed one could make a normal mouth look abnormal, or the reverse.

Illustration cropping — fixed at clinical review. The illustrations were being cut off, reported twice by the reviewing clinician. Two independent causes:

- `BoxFit.cover` on a fixed 200 px-tall box. Cover scales an image to *fill* its box and discards the overflow. The assets range from 1.09:1 to 1.50:1, so **17% to 39% of every illustration was being cropped away**, worst on the neck image at 61% visible. Nothing on screen indicated anything was missing.
- A dark gradient band carrying the site name was layered over the bottom of the image, hiding the lower part of the anatomy on every card.

Both fixed: `BoxFit.contain` inside a 280 px cap, the site name moved below the illustration, and tap-to-enlarge added with a visible "Tap to enlarge" affordance. Letterboxing an image that is whole beats a flush one that is cut.

The same `BoxFit.cover` bug affected the clinical reference gallery and, more seriously, **lesion photographs in both the patient's and the clinician's views** — where the cropped-away region is the lesion margin and surrounding mucosa a reviewer is specifically looking at. All are now `contain`. There is no remaining `BoxFit.cover` in `lib/`.

4.7 Step 3 — Lesion recording (IPO Table 6 / CareConnect module E)

Field	Control	Required	Notes
Lesion site	dropdown, 11 values	No	Lip · Inner cheek (buccal mucosa) · Gum (gingiva) · Tongue — top surface · Tongue — lateral border · Tongue — underside · Floor of mouth · Palate · Retromolar area · Neck · Other
Date first noticed	date picker	No	Range: 5 years ago to today; no future dates. Auto-fills duration if duration is still blank
Duration	numeric	No	—
Unit	dropdown	No	days · weeks (normalised to days)
Pain	switch	No	
Bleeding	switch	No	
Change in size	switch	No	

Field	Control	Required	Notes
Change in colour	switch	No	
Numbness	switch	No	
Difficulty chewing or swallowing	switch	No	
Restricted tongue or jaw movement	switch	No	
Photograph	camera / gallery	No	Triple-gated — see §4.8
Additional note	text, 500 chars	No	

A live duration hint escalates as soon as the entered duration reaches 14 days: *"That is 3 weeks. Anything lasting two weeks or longer needs a professional check."*

4.8 Photographs (IPO Table 1 / Table 6, CareConnect module E)

Capture is gated three ways: platform (`kIsWeb` disables it with an explanatory banner), photograph consent (buttons hidden), and a re-check inside the picker handler as defence in depth.

Storage went through a design change worth recording, because it affects what a clinician can see.

Problem: photographs were device-local, so a clinician reviewing the record on another device saw nothing. The obvious fix — Firebase Cloud Storage — requires the paid Blaze plan, which this pilot does not have.

Resolution: the image is stored as bytes in its own Firestore document at `assessments/{id}/lesion_photos/{lesionId}`.

This is not merely a cost workaround; it is the better fit. Cloud Storage rules cannot read a Firestore document, so they could never have checked photograph consent or which clinician the record was addressed to. In Firestore the image sits under exactly the same consent-and-recipient rules as the rest of the record.

The cost is a size limit. A Firestore document is capped at 1 MiB. Consequences, all handled explicitly:

- Capture is downscaled hard: 1024×1024 max, JPEG quality 55. Typical result 100–150 KB.
- The store refuses anything over 900 KiB (`PhotoDocumentStore.maxBytes`) rather than failing with an opaque write error.
- The patient is told twice — at capture (*"This photograph is too large to send to a doctor. Try taking it again, a little further back."*) and again on the result screen (*"Your photograph stayed on this phone... a doctor will not be able to open it."*). Silence here would leave someone believing a clinician can see an image that never left the phone.
- The same downscaled file is kept locally, so the clinician sees exactly what the patient sees.

Three photograph states are modelled distinctly, because conflating them is what caused the original bug:

Getter	Meaning
<code>hasLocalPhoto</code>	A path on the capturing device. Resolves to nothing anywhere else
<code>hasDatabasePhoto</code>	Bytes in Firestore. Current mechanism
<code>hasUploadedPhoto</code>	A Cloud Storage URL. Legacy — no longer written, still read for records created while it was in use
<code>hasRemotePhoto</code>	<code>hasDatabasePhoto</code> <code> </code> <code>hasUploadedPhoto</code> — the only forms a clinician can actually open

Every clinician-facing check uses `hasRemotePhoto`. The queue's "Photograph" pill uses it too, so the queue cannot advertise an image the record is unable to display.

4.9 Result screen (IPO Table 7)

This screen is also the commit point: the engine runs once, then `RISK_ASSESSMENT`, `SELF_EXAMINATION`, `LESION` and the photograph are written. **Nothing is shared on save** — sharing is addressed to a named clinician and only happens when the patient picks one.

Displayed, in order: headline and score band · urgency banner when applicable · "What this means" · state-specific advice bullets · "Why you got this result" (engine reasons plus the findings reported) · provisional score card with the **flag scale** (§4.9.1), per-answer contributions and any unknown answers · follow-up plan and next-check date · **Next steps** card (§4.9.2) · photograph-failure notice if applicable · send-to-doctor card · fixed no-diagnosis and provisional-scores notices.

4.9.1 Flag scale — green / yellow / red

Added at clinical review. The provisional score card shows all three bands as flags, not only the one reached, so the patient has a frame of reference for the number:

Flag	Band	Score	Action shown
 Green	Lower risk	0 to 4	Keep up prevention and check again monthly.
 Yellow	Increased risk	5 to 9	Change habits and arrange a professional check.
 Red	Higher risk	10 or more	See a dentist or doctor.

The band reached is marked with a filled flag, a tinted row and a `YOU` marker; the others are outlined. **The shape difference, not just the colour, carries the "you are here" signal**, and every row states the band name and score range in text — colour alone is unreadable to a colour-blind patient and invisible to a screen reader.

Score ranges are derived from the cut-off constants (`RiskCategory.rangeLabel`), so moving a threshold cannot leave the displayed range contradicting the engine.

The override is shown here too. A persistent red flag or a previous OSCC produces a red flag on a score of 2, and a card that displayed only the green band would contradict the instruction the same screen is giving. When an override fired, a red-flag panel states that it outranks the score band and why. A test pins this case specifically: band green, output red.

4.9.2 Next steps — referral and follow-up reminders

Added at clinical review, because both destinations already existed but were unreachable from the moment they are needed: the centres directory sat two taps into the Care & Rehab tab, and reminders behind a home-screen app-bar icon. Being told "arrange a professional oral examination" and then having to go hunting for the centre list is how a referral quietly fails.

The card sits on the result and on every past assessment, offering:

- **Find a screening centre** → the centres directory and referral-slip generator. Promoted to the primary action, and tinted with the flag colour, when the flag is red.
- **Set a follow-up reminder** → the reminders screen. The wording names the actual date — *"Be reminded two days before 29 Sep"* — rather than describing the feature, and falls back to the monthly self-check when there is no follow-up due.

4.10 Sharing (IPO §2, "With consent, send record to doctor")

Sharing is **addressed**, not broadcast. Two fields must both be set for a record to be visible: `shared_with_doctor == 1` and `shared_with_uid == <that clinician's uid>`. Switching sharing off clears the recipient, so access ends at the same instant. Changing the chosen doctor re-commits immediately, so the previous clinician loses access at the moment the new one gains it.

This was the subject of a bug worth recording: the Firestore rules permitted the read, but the client query did not filter on `shared_with_doctor`, and **Firestore rejects any query it cannot prove stays inside the permission** — rules are not filters. The record silently never appeared in the doctor's queue. Fixed with the explicit filter plus a composite index.

5. Risk and safety engine

`RiskEngine.evaluate()` — a pure function. Order of operations:

1. Derive `age_band` from the profile age and overwrite any stored value.
2. Determine `exposurePresent` — any of smoking / smokeless / areca / gutkha holding a value outside `{never, no}`.
3. Sum scores across the catalogue, in order, recording a per-variable breakdown. Skipped without scoring: conditional variables when no exposure; unanswered variables; **unknown answers (collected into `unknownAnswerKeys`)**; non-scored roles.
4. Band the total: **0–4 lower · 5–9 increased · ≥10 higher**.
5. Compute red-flag state: `neck_lump == yes` forces `persistent_neck_lump` into the flag set so it can never be dropped. `lesionPersistent` requires a red flag present **and** (duration band `≥2 weeks` **or** recorded lesion duration `≥14 days`).
6. Apply the override chain — **first match wins**:

Precedence	Condition	Output state	<code>professionalCheckRequired</code>
1	Red flag present and persistent <code>≥2 weeks</code>	<code>PROFESSIONAL CHECK REQUIRED</code>	true
2	Previous OSCC = Yes	<code>PROFESSIONAL CHECK REQUIRED</code>	true
3	Red flag present, under 2 weeks	<code>Finding recorded – review if it persists</code>	false
4	No red flag	Band result	false

7. Append supplementary reasons: previous OSCC when it co-occurs with branch 1, self-examination abnormality, unknown-answer list.
8. `referralAlert = professionalCheckRequired || category == higher || (redFlagPresent && selfExamAbnormality)`.

Red flags take precedence over the numerical score, as IPO §1 and §4 require. The band is still computed and stored even when overridden, because the validation database compares band against clinical outcome.

Note that a `higher` band alone does *not* produce `PROFESSIONAL CHECK REQUIRED` — it raises a referral alert in the clinician's queue and advises a professional examination. Only a persistent red flag or previous OSCC trips the override. This matches IPO Table 4.

5.1 Output states (IPO Table 7)

State	Headline	Storage value
<code>lowerRisk</code>	Lower risk	<code>lower_risk</code>
<code>increasedRisk</code>	Increased risk	<code>increased_risk</code>
<code>higherRisk</code>	Higher risk	<code>higher_risk</code>
<code>observeAndReview</code>	Finding recorded — review if it persists	<code>observe_and_review</code>
<code>professionalCheckRequired</code>	PROFESSIONAL CHECK REQUIRED	<code>professional_check</code>

CareConnect module D asks for four categories: "General prevention guidance / Arrange a routine dental examination / Seek professional evaluation soon / Seek urgent clinical attention". These map onto the five states above — lower risk carries prevention plus routine-examination advice, increased and higher risk carry "seek professional evaluation", and the override carries urgency. The five-state model is kept because the IPO document's `observeAndReview` case (red flag under two weeks) has no CareConnect equivalent and must not be collapsed into either neighbour.

5.2 Follow-up policy

Output state	Follow-up	Status
professionalCheckRequired	+7 days	pending
observeAndReview	lesion first noticed +14 days (or +1 day if already past)	pending
higherRisk	+14 days	pending
increasedRisk	+30 days	pending
lowerRisk	none	not required

The `observeAndReview` interval is anchored to when the patient first noticed the lesion, not to today, so the two-week question is answered about the lesion's actual age.

6. Clinician module

Maps to IPO §3 and the whole of the Clinician document. Scope is deliberately limited to review, documentation, referral and follow-up — the Clinician document's opening line, and the reason there is no automatic staging or diagnosis anywhere.

6.1 Patient queue (screen 22)

Records appear **only** after a patient addresses them to this clinician. Live consent is re-checked per case on load, so a withdrawn consent removes the record even if the flag on the assessment still says shared.

Queue priority — lower sorts first, then newest first within a bucket:

Priority	Bucket
0	Referral window lapsed with no arrival and no reminder sent yet
1	Red-flag / professional-check override
2	Referral alert
3	Everything else

Bucket 0 outranks the clinical urgency badges on purpose: a referred patient who never arrived and whom nobody has chased is the one most likely to be lost to follow-up, and it is the only state that requires a human to act rather than to read.

Three filters: Referral alerts only · Not yet reviewed · Did not attend . Three counters: referral alerts, awaiting review, did not attend.

6.2 Record review (screen 23) — IPO Table 8 left column

Read-only. Sections: app output banner · patient demographics and photo-consent state · how the app reached this output (engine reasons) · risk factors with per-answer score contributions and unknown answers · red flags reported with persistence state · patient self-examination findings · lesion records with photograph · documents the patient shared from DigiLocker · clinical assessment summary · outcome summary.

Every answer is annotated so the clinician can see how it was treated: (+4) for a contribution, (recorded, not scored) for an unweighted variable, (unknown) for a "Don't know". A clinician reading a score needs to know what is inside it.

6.3 Clinical assessment entry (screen 24)

Maps to IPO Table 8 right column and Clinician §1–§8. Tri-state controls throughout: **Yes / No / Not recorded**. Every stored boolean is nullable because "not yet decided" is a real clinical state and must not be stored as "No".

#	Field	Type	Source clause
1	Medical history	text	Clinician §1
2	Clinical examination performed	Yes/No/NR	IPO T8
3	Lesion present	Yes/No/NR	IPO T8
4	Lesion description	text	Clinician §3
5	Greatest dimension (mm)	decimal	Clinician §3
6	Impression	text	Clinician §3
7	Further investigation required	Yes/No/NR	Clinician §3
8	Biopsy required	Yes/No/NR	IPO T8

#	Field	Type	Source clause
9	Referral required	Yes/No/NR	Clinician §4
10	Referral centre	text	Clinician §4 · required when referral = Yes
11	Referral date	date	Clinician §4
12	Patient informed by text to attend a centre	Yes/No/NR	Clinician §3 (the capitalised line)
13	Date informed	date	Clinician §3
14	Patient arrived (date)	date	Clinician §5
15	Patient reminded by text	Yes/No/NR	Clinician §5
16	Date reminded	date	Clinician §5
17	Referral and follow-up instructions	text	Clinician §8 — shown to the patient
18	Follow-up date	date	Clinician §7
19	Follow-up status	text	Clinician §7

Only two save-time validations: lesion size must parse as a number, and a referral centre is required when a referral is required. Everything else is optional, because a clinician may record an examination before investigations are decided.

6.4 The two-week attendance rule (Clinician §5)

"Patient failed to arrive within two weeks" is **derived**, never typed:

```
failedToArriveWithinTwoWeeks =
  referralRequired == true
  && referralDate != null
  && patientArrivedDate == null
  && now >= referralDate + 14 days
```

Derived rather than stored so it cannot go stale — a flag written on day 14 would still read "failed to arrive" after the patient turned up on day 20. `needsAttendanceReminder` adds `patientRemindedByText != true`, and that is what drives queue bucket 0.

The entry form shows the same conclusion live while the clinician types, so the screen and the stored record cannot disagree.

6.5 Outcome entry (screen 25) — IPO §3.2, Table 9

Field	Type
Professional examination performed	Yes/No/NR
Clinical abnormality found	Yes/No/NR
Biopsy performed	Yes/No/NR
Histopathology result	text
Investigation and imaging reports	text
Final diagnosis	text
OPMD established	Yes/No/NR
OSCC established	Yes/No/NR
Referral completed	Yes/No/NR
Follow-up completed	Yes/No/NR

Professional-examination and clinical-abnormality prefill from the clinical assessment rather than asking twice.

Histopathology and final diagnosis are **free text, not enumerations**. This is deliberate: the app must not constrain a pathologist's wording, and IPO §7 requires that it "should not diagnose or stage OSCC automatically".

A live agreement preview classifies the case as the clinician types, and names the dangerous one explicitly: *"FALSE NEGATIVE: the app did not flag this patient but disease was established. This is the case type that matters most for safety."*

6.6 What the patient is not shown

`AssessmentDetailScreen` shows the patient their referral, centre, dates, attendance and their clinician's instructions. It deliberately **withholds** histopathology, clinical findings and final diagnosis, with the line *"Clinical findings and any test results are discussed with you by your clinician, not shown in the app."*

Communicating a cancer or dysplasia diagnosis is a conversation a clinician must have, not a field an app reveals.

7. Outcome / validation database

IPO §4 and Table 10. Read-only screen, computed from shared cases.

A case is **flagged** when `professionalCheckRequired || referralAlert` . It is **disease-positive** when OPMD or OSCC is established. A case with only a free-text final diagnosis counts as "outcome recorded" but is **excluded from every metric**, because the binary reference standard is the established disease status, not prose.

Metric	Formula
Sensitivity	$TP / (TP + FN)$
Specificity	$TN / (TN + FP)$
Positive predictive value	$TP / (TP + FP)$
Negative predictive value	$TN / (TN + FN)$
Positive rate per band	disease-positive / with-outcome

All return `–` rather than 0 when the denominator is zero. A rate of "0.0%" from an empty denominator would be a lie.

Also shown: data-capture counts, the 2×2 matrix, outcome by risk band (with the explicit note that a usable stratification should show the positive rate rising from lower to higher), and **a list of every false negative with a link straight into the record**.

Scope depends on role. A clinician sees only their own patients and is told so in those words: *"They are not the whole cohort, so do not read them as the pilot's performance."* A coordinator sees the whole cohort. The maths is identical; only the input list differs.

A fixed caution sits above everything: *"These are raw counts from pilot data. They have no confidence intervals, no adjustment for verification bias, and no correction for repeated assessments on the same patient. Use them to monitor the pilot, not as validation evidence."*

8. CareConnect support modules

Six modules covering CareConnect §2, §3 and modules E–J.

8.1 Education (screen 15) — CareConnect §2

Trilingual: **English, Hindi, Kannada**. 8 categories, 7 topics, 5 myths, 5 FAQs. Free-text search across titles and summaries; category filter chips.

#	Topic	Category	Content
1	What is Oral Cancer?	Overview	Understanding the disease; importance of early detection. Caution: early oral cancer is frequently completely painless
2	Major Risk Factors	Risk Factors	6 items: smokeless tobacco, areca nut/OSMF, bidi & cigarette, alcohol synergy (up to 15×), HPV-16, sharp broken teeth/dentures
3	Early Signs & Symptoms	Early Signs	The 14-day rule, 7 signs, caution against OTC gels and steroid ointments
4	Cessation & Habit Change	Prevention	Oral recovery timeline, the 4-D strategy, Quitline 1800-11-2356 , mCessation 011-22901701
5	Oral Hygiene & Nutrition	Prevention	4 protective dietary factors
6	Biopsy, Staging & Treatment	Biopsy & Care	What a biopsy is; explicitly rebuts the "biopsy spreads cancer" myth
7	Caregiver & Family Guidance	Caregivers	Supporting without blaming

Myths covered: age/smoking-only misconception · biopsy spreads cancer · painless means benign · plain supari is safe · oral cancer is contagious.

FAQs: self-examination frequency · normal vs cancerous ulcer · can leukoplakia resolve · what to do on "Professional Check Required" · does the app diagnose (answer: "No. The app strictly supports awareness, self-examination, and structured referral.").

Static content, no persistence.

8.2 Clinical reference photo gallery (screen 14) — CareConnect §3 ("pics of red, white lesion")

Six pinch-to-zoom reference photographs with key signs and clinical significance, reachable from four places including a per-site button on the self-examination screen:

Asset	Condition
A	Homogeneous Leukoplakia (White Patch)
B	Erythroleukoplakia (Mixed Red & White Patch)
C	Verrucous Leukoplakia (Thickened White Lesion)
D	Chronic Non-Healing Oral Ulcer
E	Exophytic Lump / Oral Carcinoma
F	Oral Submucous Fibrosis (OSMF)

Header states: "photographs cannot diagnose cancer. Only a clinician and biopsy can provide an accurate diagnosis."

8.3 DigiLocker (screen 16) — CareConnect module G

All 13 document categories from module G:

#	Category	#	Category
1	Consultation Record	8	Prescription
2	Clinical Photograph	9	Surgery / Radiotherapy
3	Biopsy & Histopathology	10	Chemotherapy Record
4	Blood Investigation	11	Discharge Summary
5	Imaging (CT/MRI/OPG)	12	Follow-up Note
6	Diagnosis & Staging	13	Bills & Insurance
7	Treatment Plan		

Per record: category (required), title (required), doctor/hospital, date of report (no future dates), notes, optional attachment, and a **per-document sharing switch**.

Module G's *"Users should control which records are shared and with whom"* is implemented literally: each document carries its own `is_shared` flag, and the clinician's record screen reads only shared documents. The query filter is mandatory, not an optimisation — the rules permit a clinician to read only shared documents, and Firestore rejects a query it cannot prove stays inside that permission.

Attached pictures are stored server-side. The image is held as bytes at `users/{uid}/digilocker_files/{recordId}` — the same mechanism as lesion photographs (§4.8), sharing one implementation and one size limit via `ImageDocumentStore`.

This was a defect until clinical review. The locker previously kept only `local_file_path`, a path on the capturing device, which meant a filed biopsy report or clinical photograph disappeared on any other device, was lost with the phone — precisely what a records locker exists to prevent — and showed the reviewing clinician nothing even when the patient had marked it shared.

Design notes:

- The bytes live in a sibling document, not on the record, so listing the locker does not download every image.
- The file document id **is** the record id. That makes the pairing unambiguous and lets the rules locate the parent record.
- The file document deliberately carries **no sharing flag of its own**. The patient makes one `is_shared` decision, on the record, and the rules read that record to authorise the image — so the two cannot drift apart.
- Capture is downscaled to 1400 px at quality 60 and refused above ~900 KiB, with the reason given.
- The image is written **before** the record, so `has_image` on the record is a statement of fact rather than a hope. If the image fails, the record still saves and the patient is told it will not be visible on another device.
- Records filed before this existed show an explanatory note rather than an empty space, so nobody assumes an old picture is safe in the locker.

Still not built: multi-page PDF reports. The 1 MiB per-document ceiling makes them impractical without Cloud Storage.

8.4 Cessation (screen 17) — CareConnect module H

Module H requirement	Implementation
Tobacco and areca-nut cessation resources	Quitline 1800-11-2356 (8 AM-8 PM), mCessation missed-call 011-22901701, plus education topic 4
Quit-date setting	Date picker, up to 30 days ahead
Habit tracking	Live streak in days and hours; money saved (₹); units avoided

Module H requirement	Implementation
Daily motivational messages	Built, limited — static motivational copy and milestone feedback. No scheduled daily message engine
Identification of triggers	7-option trigger dropdown with 1–5 intensity slider, logged per craving
Referral to cessation services	Quitline and mCessation cards
Progress monitoring	6 recovery milestones, achieved state computed from elapsed time
Relapse-support information	Fixed card: "A slip is not a failure; it is data on what trigger was challenging."

Quit plan fields: quit date · habit types (Smokeless Tobacco/Gutkha, Bidi/Cigarette, Areca Nut/Paan/Supari, Alcohol) · units per day · cost per unit · craving log.

Craving triggers: Work/Stress · Social gathering/Friends · After eating a meal · Boredom/Free time · With Chai/Coffee · Anxiety/Restlessness · Other.

Recovery milestones: 20 minutes (heart rate) · 24 hours (carbon monoxide) · 48 hours (taste and smell) · 2 weeks (oral mucosa regeneration) · 1 month (gums and teeth) · 1 year (oral cancer risk drops by 50%).

Craving assistance uses the 4-D strategy: Delay · Deep breath · Drink water · Distract.

Stored at `users/{uid}/cessation/plan`. Not visible to clinicians.

8.5 Appointments and referral (screen 19) — CareConnect module F

Four centre types: Dental Institute / College · Comprehensive Cancer Centre · Oral Medicine & OMFS Dept · Community Health Centre.

Five seed centres, each with name, type, city, state, full address, phone, timings and a service list:

Centre	Type	City
Coorg Institute of Dental Sciences (CIDS)	Dental institute	Coorg (Virajpet)
Kidwai Memorial Institute of Oncology	Oncology centre	Bengaluru
Tata Memorial Centre (TMC)	Oncology centre	Mumbai
Government Dental College & Research Institute (GDCRI)	Dental institute	Bengaluru
Dr. B.R. Ambedkar IRCH (AIIMS)	Oncology centre	New Delhi

Centres are served from Firestore `screening_centers` and seeded with `tools/seed_centers.mjs`; the bundled list is only a fallback. Clinic phone numbers and timings change, and a hardcoded list cannot be corrected without shipping a release. The banner wording changes to match the source: the bundled fallback says "These details are the copy built into the app and may be out of date."

Appointment requests — Built, limited, and the limit is stated everywhere it could mislead. The app has no booking integration with any centre. A request is the patient's own record of an intention to attend. It starts as `Saved — centre not yet contacted`, and the patient advances it to `Confirmed with the centre` then `Attended` once they have actually phoned.

Wording used, verbatim: - "This saves a reminder for you. The app cannot book for you — you still need to phone the centre." - "Saved to your record for 21 Sep. Now call +91 8274 256479 to confirm — the centre has not been contacted automatically." - "Saved by you. The app does not contact centres for you."

A save failure is surfaced, not swallowed, so the patient is never shown a false confirmation and left waiting for a call that never comes.

Referral letters (module F) are generated as a plain-text clinical referral slip containing demographics, provisional risk band, urgency status, red-flag trigger, lesion and symptoms, four recommended clinical actions, and a safety disclaimer: *"It does NOT constitute a confirmed diagnosis or staging of Oral Squamous Cell Carcinoma."* Copyable to clipboard to hand to a clinician.

Not built: teleconsultation (module F, "where legally and technically appropriate") — requires a regulatory decision, not a code change. `Call` buttons display the number for manual dialling rather than launching the dialer.

8.6 Post-treatment rehabilitation (screen 20) — CareConnect §9 target users, treatment support

Three treatment modalities, selected by the patient:

Modality	Hydration interval	Exercises	Focus
Surgery Only	3 h	3	Jaw opening physiotherapy, tongue speech drills, scar mobility, neck/shoulder after node dissection
Surgery + Radiotherapy	2 h	2	Xerostomia, saliva substitutes, radiation-carries prevention, swallow reflex
Surgery + Chemo + Radiotherapy	2 h	2	Neutropenia infection barrier, oral mucositis, nutrition, psychological support

7 exercises total, each with target area, instruction, frequency and clinical rationale. 9 clinical priorities and 13 special instructions across the three protocols, including a febrile-neutropenia escalation (*"Any fever above 100.4°F (38°C) is a medical emergency"*), the dry-mouth protocol with sodium-bicarbonate rinse recipe and 1.1% neutral sodium fluoride gel, and explicit psychological-support guidance.

A live hydration prompt turns due based on the modality's interval.

Built, limited: nothing on this screen is persisted. Modality choice, hydration time and completed exercises are session-only, and the hydration prompt is an in-screen colour change, not a scheduled notification.

8.7 Emergency guidance (screen 21) — CareConnect module J

All five module J situations covered, plus one:

#	Condition	Module J clause
1	Severe / Uncontrolled Oral Bleeding	Severe bleeding
2	Difficulty Breathing or Noisy Breathing (Stridor)	Difficulty breathing
3	Rapidly Increasing Swelling in Neck or Face	Rapidly increasing swelling
4	Inability to Swallow Saliva or Fluids	Difficulty swallowing
5	High Fever (>100.4°F / 38°C) During Chemotherapy	High fever during cancer treatment
6	Severe Post-Operative Complications	Severe post-operative complications

Each carries an immediate first step and an explicit escalation threshold — e.g. bleeding persisting past 20 minutes of firm pressure, inability to swallow for more than 6 hours, any dark or blue flap discolouration.

Helplines: `112` (National Emergency) and `108` (National Ambulance).

Built, limited: the buttons display the number in a dialog rather than launching the dialer. Labelled "Call 112", which implies dialling — worth changing to "Show emergency number" or wiring `url_launcher`.

Reachable in one tap from the top of the Check Hub.

9. Notifications and reminders — CareConnect module I

All eight module I reminder types exist as scheduling kinds:

Kind	Notification title	Module I clause
selfExamination	Time for your mouth self-check	Monthly self-examination
dentalCheckUp	Dental check-up due	Dental check-ups
followUpVisit	Follow-up visit due	Follow-up visits
biopsyAppointment	Biopsy appointment	Biopsy appointments
medication	Medication reminder	Medication schedules
investigation	Investigation due	Investigation dates
treatmentSession	Treatment session	Treatment sessions
cessationMilestone	Quit-plan milestone	Tobacco-cessation milestones

Schedule: monthly self-check at **09:00** on the same day next month; date-anchored reminders fire at 09:00, **2 days before** the due date. Channel `oralcare_reminders`, timezone `Asia/Kolkata`, inexact scheduling (no exact-alarm permission is requested, so no special permission is needed).

Module I requires *"Notifications must be optional"*: nothing is scheduled until the patient taps **Set reminders**, and **Turn all reminders off** clears everything. Privacy: *"Reminders stay on this device. Nothing is sent to anyone else."*

Built, limited — Android only. The browser cannot schedule them, and the screen says so rather than pretending: *"This browser cannot show scheduled reminders. Install the Android app to receive them. Your due dates are still listed below."* When permission is declined it says that too. A reminders screen that silently fails is worse than one that admits it.

10. Language — CareConnect §2 and §11, Questionnaire

Two layers, on purpose.

Clinical questions: English and Kannada. Every Kannada term is taken verbatim from the Sree questionnaire — gender options, yes/no/don't-know, habits, sites, all 8 examination sites and all 13 red flags. In Kannada mode the English term stays visible underneath, because that is the term a clinician will ask about.

Education module: English, Hindi and Kannada.

Hindi is deliberately not offered for clinical questions. The source documents contain no verified Hindi for them. Machine-translating a clinical red-flag term into a language no clinician on the project has reviewed is a patient-safety risk, not a feature. Tests enforce that every red flag and every examination site has a Kannada term and that no term is left as a placeholder.

Language choice is session-scoped and never written to the profile, so a guest can use Kannada without an account.

Built, limited: the surrounding interface chrome (buttons, headings, navigation) is still English. Full UI localisation needs `flutter_localizations` and a reviewed translation of every string.

11. Data model – IPO §6, Table 11

IPO §6 specifies six tables with `Patient_ID` as the primary linking key. All six exist.

IPO table	Firestore location
USER	<code>users/{uid}</code>
RISK_ASSESSMENT	<code>assessments/{assessmentId}</code>
SELF_EXAMINATION	field <code>self_exam</code> inside the assessment document
LESION	<code>assessments/{assessmentId}/lesions/{lesionId}</code>
CLINICIAN_ASSESSMENT	<code>assessments/{assessmentId}/clinical/current</code>
OUTCOME	<code>assessments/{assessmentId}/outcome/current</code>

Self-examination is stored inline rather than as a subcollection because it is strictly one-per-assessment; a subcollection would add a read for no benefit.

11.1 Full path list

Path	Contents
<code>users/{uid}</code>	Profile, role, consent flags
<code>users/{uid}/digilocker_records/{id}</code>	Patient documents, with <code>is_shared</code> and <code>has_image</code>
<code>users/{uid}/digilocker_files/{id}</code>	Attached picture bytes. Id matches the record id
<code>users/{uid}/cessation/plan</code>	Quit plan and craving log
<code>users/{uid}/appointments/{id}</code>	Patient-authored visit intentions
<code>patient_ids/{patientId}</code>	Uniqueness reservation. Doc id <i>is</i> the Patient ID
<code>doctors/{uid}</code>	Public clinician directory. No patient data, no private contact details
<code>screening_centers/{id}</code>	Centre directory, Admin-SDK seeded
<code>assessments/{id}</code>	Risk assessment + inline self-examination + sharing + follow-up
<code>assessments/{id}/lesions/{id}</code>	Lesion records
<code>assessments/{id}/lesion_photos/{id}</code>	Photograph bytes (Blob), content type, byte count
<code>assessments/{id}/clinical/current</code>	Clinician assessment
<code>assessments/{id}/outcome/current</code>	Outcome / reference standard

The clinician directory is a separate collection from `users` on purpose: patients must be able to list clinicians without being able to read any profile document.

11.2 RISK_ASSESSMENT fields – IPO Table 11 "*Risk factors, individual values, total score, category*"

`patient_id` · `owner_uid` · `created_at` · `answers_json` (every individual value) · `risk_score` · `risk_category` · `red_flag` · `red_flags_json` · `lesion_persistent` · `previous_oscc` · `professional_check` · `referral_alert` · `output_state` · `unknown_keys_json` · `breakdown_json` (per-variable contribution) · `reasons_json` · `shared_with_doctor` · `shared_with_uid` · `follow_up_due` · `follow_up_status`

The full engine output is persisted, not recomputed on read. A historical record must always show the score and category that were actually produced at the time — re-running a changed algorithm over old data would corrupt the validation set.

`breakdown_json` and `unknown_keys_json` exist so an auditor can reconstruct exactly how a total was reached.

12. Security — CareConnect §11

CareConnect §11 requirement	Status
Encrypted data transmission and storage	Built — Firebase TLS in transit, encryption at rest
Secure login	Built — Firebase Authentication; passwords never reach Firestore
Optional multi-factor authentication	Not built — needs a Firebase Auth MFA configuration decision
Role-based access control	Built, limited — see §2.1
Consent-based record sharing	Built — enforced in rules, not just in UI
Audit logs for access and modifications	Not built — needs Cloud Functions (paid plan) or a rules-level logging sink
Regular data backup	Not built — Firestore scheduled export requires the paid plan
Data recovery after device loss	Built — records are server-side; sign in on a new device restores them. Exception: device-local photographs and DigiLocker attachments
Secure logout from shared devices	Built
Minimal PII collection	Built — the clinician's queue shows Patient ID, age and gender only; never a name
Clear privacy and retention policy	Not built — a document, not code
Multilingual and accessibility-friendly	Built, limited — see §10

12.1 Firestore rules summary

Path	Read	Write
users/{uid}	Self, or any doctor	Self only. Delete denied
users/{uid}/digilocker_records/{id}	Self, or doctor when <code>is_shared == 1</code>	Self only
users/{uid}/digilocker_files/{id}	Self, or doctor when the parent record has <code>is_shared == 1</code>	Self only
users/{uid}/cessation , .../appointments	Self only	Self only
doctors/{uid}	Any signed-in user	That clinician only. Delete denied
screening_centers/{id}	Any signed-in user	Denied — Admin SDK only
patient_ids/{id}	<code>get</code> only; <code>list</code> denied	Create only, and only with own uid. Update and delete denied
assessments/{id}	Owner, or the addressed doctor, or a coordinator when shared	Owner only, and <code>owner_uid</code> must equal the caller
.../lesions/{id}	Same as parent	Parent owner only
.../lesion_photos/{id}	Same as parent	Parent owner only
.../clinical/{id}	Parent owner, addressed doctor, or coordinator	Addressed doctor only — a coordinator can read, never write
.../outcome/{id}	Parent owner or addressed doctor	Addressed doctor only
everything else	Denied	Denied

Subcollection rules were the source of a real data-loss bug: **Firestore rules do not cascade**. A `users/{uid}` rule does not cover `users/{uid}/digilocker_records`, so those writes were being silently rejected while the UI reported success. Every subcollection now has an explicit block.

Note an asymmetry left in place: `clinical` grants a coordinator read access, `outcome` does not. Since the outcome document is the validation reference standard, this looks like an omission worth reviewing — a coordinator computing cohort statistics reads outcomes through the same query.

13. Clinical and ethical safeguards — CareConnect §12

Safeguard	Implementation
Clear disclaimer that the app does not diagnose	<code>ClinicalNotices.noDiagnosis</code> , shown on role selection, consent, every result and every history detail
No treatment recommendations without clinical evaluation	Advice is confined to prevention, cessation and "see a professional". No drug, dose or procedure is ever suggested
No interpretation of photographs as diagnosis	The gallery states photographs cannot diagnose; the clinician's view labels every image <i>"Patient-captured image. Lighting, focus and angle are uncontrolled, so it does not replace direct examination."</i>
Referral for suspicious findings	The override chain in §5, the referral alert in the queue, and the centres directory
Informed consent for collection and sharing	Three-gate consent screen, re-editable, enforced in the router and the rules
Special safeguards for minors and vulnerable users	Not built. Age is captured (0–120) but there is no minor-specific pathway, guardian consent, or restriction. Needs a clinical and legal decision before it can be coded
Ethical approval for research data	Process, not code
De-identification for research	Built, limited — the clinician's queue is already name-free. There is no export pipeline, so no de-identification step exists to build yet
Transparent explanation of any AI feature	There is no AI or ML anywhere in this app. The risk engine is a transparent additive score with a published weight table and a per-answer breakdown shown to both patient and clinician
Human clinical oversight of automated alerts	Every alert routes to a clinician who must record their own examination. <code>doctorResponsibility</code> notice: <i>"App output is decision support only."</i>
Mechanism for reporting inaccurate information	Not built. No in-app feedback channel exists

Two further safeguards not requested but present:

- **provisionalScores** on every result and history screen: *"Scores and thresholds in this build are provisional for pilot development and have not yet been clinically validated."* IPO Table 3 marks the cut-offs provisional; the app repeats it to the patient rather than only in a document.
- **validationCaveat** above the validation statistics, stating they are not validation evidence.

13.1 Consent wording accuracy — corrected

Three strings on or above the consent form had gone stale when records moved from device-local storage into Firestore. All three are now corrected, and a test file (`test/consent_wording_test.dart` , 12 tests) pins the claims so they cannot drift back.

String	Was	Now
<code>storageNotice</code>	<i>"Records are stored on this device only. Nothing is uploaded."</i> — false since the Firestore migration	States records are saved to the patient's account, encrypted in transit and at rest, readable only by them unless shared. Also discloses what the pilot lacks: no audit trail of who opened a record, no automatic backup
<code>consentPhotoDetail</code>	<i>"Photographs stay on this device unless I also agree to share my record with a doctor"</i> — false; the photograph is written to Firestore as soon as photograph consent is given, independent of share consent	States the photograph is saved to the patient's own account so it is available on any device they sign in to, and that no doctor can open it unless they also share that record
<code>consentShareDetail</code>	<i>"a doctor using this app may see my record"</i> — implies any clinician; sharing is addressed to one	States that only the chosen doctor can open it, that it can be withdrawn at any time, and discloses the pilot coordinator , who can read shared records cohort-wide for validation

The coordinator disclosure was an omission, not a drift: the role was added after the consent text was written, and a patient cannot consent to a disclosure they were never told about.

A related screen string was corrected with them: the lesion screen's photograph section said *"Stays on this device unless you share your record with a doctor."*

Design decision recorded. There were two ways to resolve the photograph contradiction: change the text to match the behaviour, or gate the upload on share consent so the original promise became true. The text was changed. Holding the photograph in the patient's own account is a benefit to them — it survives device loss and appears in their history on any device they sign in to — and the Firestore rules already restrict it to the owner plus any clinician they addressed the record to. So "saved to your account, no doctor can see it unless you share" is both accurate and the more useful behaviour. Gating the upload would have traded a real patient benefit for wording convenience.

14. Traceability matrices

14.1 IPO document — `Oral_Cancer_Mobile_App_Patient_Doctor_Login.docx`

Clause	Requirement	Status	Where
§1	Four-layer architecture	Built	§1
§2	Patient flow order	Built	§4.1
§2.1 T1	Patient ID / permitted identifier	Built	§4.2
§2.1 T1	Age as demographic + risk variable	Built, required	§4.2, §4.4
§2.1 T1	Sex, where used	Built as <i>Gender</i> , required	§4.2
§2.1 T1	Consent for app/self-exam — proceed only when Yes	Built, router-enforced	§3.2, §4.3
§2.1 T1	Consent for photograph	Built	§4.3, §4.8
§2.1 T1	Consent to share with doctor	Built	§4.3, §4.10
§2.2 T2	All 13 risk variables with provisional scores	Built	§4.4
§2.2 T2	Gutkha: capture exposure, weight to be validated	Built as unweighted, labelled	§4.4
§2.2 T2	Previous OPMD "Don't know" retained as unknown	Built, never scored as zero	§4.4
§2.2 T3	Bands 0–4 / 5–9 / ≥10	Built	§5
§2.2	"Scores and cut-offs are provisional"	Built, surfaced to the patient	§13
§2.3	10 red-flag questions	Built, extended to 13	§4.5
§2.3 T4	No red flag → numerical category	Built, branch 4	§5
§2.3 T4	Red flag <2 weeks → record, advise observation	Built, branch 3	§5
§2.3 T4	Red flag ≥2 weeks → PROFESSIONAL CHECK REQUIRED	Built, branch 1	§5
§2.3 T4	Previous OSCC → follow-up irrespective of score	Built, branch 2	§5
§2.4 T5	7 self-examination sites, examined + abnormality	Built, extended to 8	§4.6
§2.4	Abnormality opens lesion module	Built	§4.1
§2.5 T6	All 12 lesion fields	Built	§4.7
§2.5 T6	Photograph, optional upload with consent	Built	§4.8
§2.6 T7	All four output states with guidance	Built as 5 states	§5.1
§3	Doctor flow order	Built	§6
§3.1 T8	Everything the doctor sees	Built	§6.2
§3.1 T8	Everything the doctor enters	Built, 19 fields	§6.3
§3.2 T9	All 13 outcome backend fields	Built	§6.5, §11.2
§4	Combined decision logic, red flag over score	Built, precedence chain	§5
§5	Input → process → output map	Built	§1
§6 T11	Six tables, Patient_ID linking key	Built	§11
§7	Two separated authenticated interfaces	Built	§2, §3.2
§7	Must not diagnose or stage automatically	Built — free-text diagnosis, no AI	§6.5, §13

IPO coverage: complete.

14.2 Clinician document — Copy 3.docx

Clause	Requirement	Status	Where
Scope	Limited to review, documentation, referral, follow-up	Built	§6
§1	Patient ID, age, sex	Built	§6.2
§1	Relevant medical history	Built — clinician-entered field 1, plus the patient's own from their profile	§6.3
§1	Tobacco/areca/alcohol history	Built — risk factors with score contributions	§6.2
§1	Previous oral lesion or OSCC history	Built	§6.2
§2	Reported symptoms and duration	Built	§6.2
§2	Lesion site	Built	§6.2
§2	Self-examination findings	Built	§6.2
§2	Patient-uploaded photograph	Built	§4.8, §6.2
§2	Preliminary risk-assessment result	Built , with engine reasoning	§6.2
§3	Lesion description and size	Built , fields 4–5	§6.3
§3	Clinical findings	Built	§6.3
§3	Provisional clinical impression	Built , free text	§6.3
§3	Further investigation required Yes/No	Built , tri-state	§6.3
§3	PATIENT INFORMED BY TEXT REPLY TO ATTEND A CENTRE	Built, limited — recorded with date. The app sends no SMS; it records that the clinician did	§6.3
§4	Referral required Yes/No	Built	§6.3
§4	Referral centre/clinician	Built , required when referral = Yes	§6.3
§4	Referral date	Built	§6.3
§5	Patient arrived (date)	Built	§6.3
§5	Patient failed to arrive within two weeks	Built , derived not typed; drives queue priority 0	§6.4
§5	Patient reminded by text	Built , with date	§6.3
§6	Biopsy/histopathology report	Built	§6.5
§6	Relevant investigation/imaging reports	Built	§6.5
§6	Final diagnosis, where available	Built	§6.5
§7	Follow-up date	Built	§6.3
§7	Follow-up status	Built , free text	§6.3
§8	Share selected patient information with authorised clinicians	Built — addressed sharing, per-document DigiLocker sharing	§4.10, §8.3
§8	Send referral/follow-up instructions to the patient	Built — field 17, surfaced in the patient's own record	§6.3, §6.6

Clinician coverage: complete. One constraint: the app records that the patient was informed or reminded by text; it does not send the SMS. No messaging gateway is integrated.

14.3 CareConnect — Final cHeck.docx

Clause	Requirement	Status	Where
§1	Name, age, gender	Built , all required	§4.2
§1	Contact details	Built — mobile, email	§4.2
§1	Location	Built — city, PIN	§4.2
§1	Relevant medical history	Built	§4.2
§1	Tobacco, alcohol, areca-nut use	Built — risk questionnaire	§4.4
§1	Previous oral lesions / cancer history	Built	§4.4
§1	Allergies and current medications	Built — profile screen	§3 #6
§1	Emergency contact information	Built	§4.2
§1	Guest-information mode	Built	§2
§2	All 13 education content areas	Built — 7 topics, 5 myths, 5 FAQs	§8.1
§2	English and regional Indian languages	Built — EN/HI/KN education, EN/KN clinical	§10
§3	7 self-examination sites incl. back of throat	Built — 8 sites	§4.6
§3	Illustrations, instructions, reminders	Built, limited — 7 of 8 illustrations exist	§4.6, §15
§3	"pics of red, white lesion"	Built — 6-photo clinical gallery	§8.2
§3	All 10 symptom questions	Built — 13 red flags	§4.5
§3	Must state it is not a substitute for clinical examination	Built	§13
§D	Risk questionnaire: 10 factor groups	Built	§4.4
§D	Family history	Recorded only — no agreed weight	§4.4
§D	Immunosuppression	Recorded only — no agreed weight	§4.4
§D	Four output categories	Built as 5 states	§5.1
§E	Lesion documentation, all 7 fields	Built	§4.7
§E	Clinical photographs where possible	Built, limited — ≤900 KiB	§4.8
§E	Consultation details, follow-up dates	Built — clinician review shown in the patient's record	§6.6
§F	Dental institutes / specialist search	Built — 5 centres, 4 types, searchable	§8.5
§F	Location-based screening-centre information	Built, limited — city filter, not GPS-based	§8.5
§F	Appointment requests	Built, limited — patient-authored, no booking integration	§8.5
§F	Referral letters	Built — generated clinical slip	§8.5
§F	Follow-up reminders	Built	§9
§F	Emergency contact information	Built	§8.7
§F	Teleconsultation support	Not built — regulatory decision required	§8.5
§F	Clinician availability verified and updated	Built — Firestore-served, seedable without a release	§8.5
§G	All 13 document categories	Built	§8.3
§G	Users control which records are shared and with whom	Built — per-document flag, rule-enforced	§8.3
§H	All 8 cessation features	Built, limited — no scheduled daily message engine	§8.4
§H	Avoid blame, encourage professional support	Built — relapse card, quitline referral	§8.4

Clause	Requirement	Status	Where
§I	All 8 reminder types	Built, limited — Android only	§9
§I	Notifications optional, privacy addressed	Built	§9
§J	All 5 emergency situations	Built — 6 conditions	§8.7
§8	Proposed 12-step user workflow	Built — every step has a screen	§4.1, §8
§9	Target users	Design input, not a feature	—
§10	Clinician module, 9 capabilities	Built, limited — see below	§6
§10	Role-based access	Built, limited	§2.1
§11	Technology and security, 11 items	Partial — 4 absent, itemised	§12
§12	Clinical and ethical safeguards, 11 items	Partial — 2 absent, itemised	§13

CareConnect §10 clinician capabilities in detail:

§10 capability	Status
Register patients with consent	Not built — patients self-register. Clinician-initiated enrolment does not exist
Review patient-submitted information	Built
Record clinical findings	Built
Upload reports and photographs	Built, limited — free-text report fields; no clinician file upload
Generate referrals	Built — referral fields; the printable slip is patient-side
Recommend follow-up	Built
Communicate securely with patients	Built, limited — one-way written instructions surfaced in the patient's record. No messaging thread
Monitor high-risk patients	Built — priority queue, filters, attendance tracking
Create anonymised datasets for approved research	Not built — no export pipeline

14.4 Questionnaire — sree cancer questionnaire final (1).docx

Used as the **terminology source**, not as a screen to reproduce. The questionnaire is a knowledge–attitude–practice survey instrument (sections 1–4: awareness, causes, features, information sources) and the app is a risk-assessment tool; they serve different purposes.

Questionnaire element	Use in the app
GENDER / ಛರ — Male / Female / Transgender	Gender options and their Kannada terms
Bilingual Agree / Disagree / Don't know	ClinicalTerms answer terms
§2 causes: smoking, smokeless tobacco/pan, alcohol, betel leaf & areca, family history	Kannada terms for the habit variables
§3 features: white patch, red patch, non-healing ulcer, excess tissue growth, swelling, continuous pain, pus/boil, reduced mouth opening, bleeding gums, sudden loose teeth	Kannada terms for the red-flag checklist; loose_teeth red flag added from §3.10
Sites: mouth, throat	Kannada terms for examination sites; throat site added
§3.3 / §3.11 three-week ulcer threshold	Discrepancy — see §15

The questionnaire's demographic fields not carried into the app — education level, place of residence, annual income, occupation — are research-survey variables with no scoring role in IPO Table 2. They were not added because collecting income and occupation without a scoring or reporting purpose conflicts with CareConnect §11's "minimal collection of personally identifiable information".

15. Open items

15.1 Clinical decision required: 2 weeks or 3 weeks?

The app escalates a persistent lesion at ≥ 14 days, per IPO §2.2 ("Duration of lesion: <2 weeks / ≥ 2 weeks — ≥ 2 weeks = referral trigger"), IPO Table 4, and IPO §4.

The Sree questionnaire says **three weeks** twice: - §3.3 "Non-healing Wound or Ulcer of more than three weeks" - §3.11 "I have to see the dentist if a mouth ulcer lasts for more than three weeks"

These cannot both be right. The threshold appears in the engine, the follow-up policy, the clinician's attendance window, the duration hint, the education content ("The 14-Day Rule") and multiple tests, and it is defined once as

`RiskCatalog.persistenceThresholdDays = 14`, so changing it is a one-line change plus test updates.

This needs the clinical team to choose. Two weeks is the more cautious threshold and refers more patients; three weeks matches the questionnaire the team wrote. The app currently follows the IPO document because that is the specification it was built against.

15.2 Images still needed

Asset	For	Status
~~ assets/images/8.png ~~	Self-examination site: back of the throat	Added — see §4.6
Red patch alone	Sign reference	Suggested addition
Swelling of mouth or jaw	Sign reference (Questionnaire §3.5)	Suggested addition
Pus / boil	Sign reference (Questionnaire §3.7)	Suggested addition
Bleeding gums	Sign reference (Questionnaire §3.9)	Suggested addition
Loose teeth	Sign reference (Questionnaire §3.10)	Suggested addition
Normal healthy mouth	Baseline for comparison	Suggested addition

The existing gallery (A–F) covers leukoplakia, erythroleukoplakia, verrucous leukoplakia, chronic ulcer, exophytic lump and OSMF. The gaps are the questionnaire's §3 features that have no reference image, plus a normal baseline — patients consistently benefit more from "this is what normal looks like" than from another pathology photograph.

15.3 Weights awaiting clinical agreement

`gutkha_pan_masala`, `family_history` and `immunosuppression` are recorded but score zero. A test enforces that they do not move a patient's score. Supply weights and the test is the only thing that needs changing.

15.4 Correctness items to fix

Item	Severity
~~Consent wording claimed records and photographs stayed on the device~~	Fixed — see §13.1
Enrolment code + <code>hasDoctorProfile()</code> rules fallback allow self-assigned clinician role	High — must be removed before real deployment
<code>outcome</code> rules block has no coordinator read branch while <code>clinical</code> does	Medium — review whether intentional
Emergency and centre "Call" buttons display a number rather than dialling	Low — but the label implies dialling
~~DigiLocker attachments are device-local paths~~	Fixed — see §8.3
~~Illustrations and photographs cropped by <code>BoxFit.cover</code> ~~	Fixed — see §4.6

Item	Severity
Rehabilitation screen state is not persisted	Low

15.5 Not built, with reasons

Item	Source	Reason
Teleconsultation	CareConnect §F	Regulatory decision required
Audit logs	CareConnect §11	Needs Cloud Functions (paid plan)
Scheduled backups	CareConnect §11	Firestore export needs the paid plan
Multi-factor authentication	CareConnect §11	Configuration decision
Minor / vulnerable-user safeguards	CareConnect §12	Needs a clinical and legal pathway definition first
Anonymised research export	CareConnect §10, §12	No export pipeline
In-app inaccuracy reporting	CareConnect §12	Not started
Clinician-initiated patient registration	CareConnect §10	Not started
iOS build	CareConnect §11	Needs a macOS signing identity and an Apple developer account
Full UI localisation	CareConnect §11	Needs reviewed translations of all interface strings

16. Test coverage

174 tests across 13 files. All pass.

File	Covers
risk_engine_test.dart	Every variable's score contribution, band cut-offs, override precedence, unknown-answer retention, unweighted variables not moving the score, the 14-day boundary at 13 / 14 / 20 days, neck-lump handling, self-exam interaction
sharing_test.dart	Addressed sharing invariants the Firestore rules depend on — flag alone and recipient alone must both fail
attendance_test.dart	The two-week attendance rule, including the day-14 boundary and the reminder-needed state
registration_profile_test.dart	Required age and gender, legacy sex key fallback
screening_centers_test.dart	Firestore-vs-bundled source selection
translation_test.dart	Every red flag and exam site has a Kannada term; no placeholders; gender options match the questionnaire
reminder_service_test.dart	Platform gating, id allocation
lesion_photo_test.dart	The three photograph states, legacy-row compatibility, the 1 MiB limit, and that withdrawing consent cuts every route to the image
consent_wording_test.dart	Consent-form accuracy: the stale device-only claims cannot reappear, and the account/encryption/audit-gap/coordinator/withdrawal disclosures must stay stated
review_feedback_test.dart	The clinical-review changes: flag-to-band mapping, that an override shows red over a green band, band ranges derived from the cut-offs, and that a DigiLocker local path is not a viewable image
enrolment_switch_test.dart	Behaviour under both ALLOW_ENROLMENT_CODE builds
new_features_test.dart	CareConnect module models
app_smoke_test.dart	App boots, role selection renders

Run: flutter test and flutter test --dart-define=ALLOW_ENROLMENT_CODE=false .

16.1 What the tests do not cover

No integration or end-to-end test exercises a real Firebase backend. In particular the following have been verified by build and by rules compilation, **not** by running the actual flow across two devices:

- A photograph captured on one device opening in the clinician's view on another
- The Firestore rules rejecting an unauthorised read at runtime
- Notification delivery on a physical Android device

These need manual verification before demonstration. The README carries a seven-step two-device walkthrough for exactly this.

17. Build and deploy

```
flutter analyze lib test
flutter test
flutter build apk --release
flutter build web -t lib/main.dart
firebase deploy --only firestore:rules,hosting --project oral-cancer-pilot-1027-dc3a3
```

Regenerate this document as a PDF after editing it:

```
python3 -m pip install markdown      # once
python3 tools/make_spec_pdf.py       # writes SPECIFICATION.pdf
```

Provisioning tools (require a service-account key):

```
node tools/grant_doctor.mjs --username <name>           # grant the doctor claim
node tools/grant_doctor.mjs --coordinator --username <name> # grant the coordinator claim
node tools/seed_centers.mjs                             # seed the centres directory
```

Build switches:

Switch	Default	Effect
ALLOW_ENROLMENT_CODE	true	false removes clinician self-registration entirely
ENROLMENT_CODE	ORAL-PILOT-2026	Change the code without editing source

Appendix A — Content inventory totals

Item	Count
Screens	26
Risk-assessment variables	15 (max score 27)
Red-flag checklist items	13
Self-examination sites	8
Lesion site options	11
Lesion symptom switches	7
Patient output states	5
Clinician assessment fields	19
Outcome fields	10
Validation metrics	4 + per-band positive rate
Education topics	7 (3 languages)
Myths and facts	5
FAQs	5
Clinical reference photographs	6
DigiLocker categories	13
Screening centres (seed)	5, across 4 types
Cessation milestones	6
Craving triggers	7
Rehabilitation modalities	3 (7 exercises, 13 special instructions)
Emergency conditions	6
Reminder types	8
Firestore collections / subcollections	13
Tests	174

Appendix B — Key constants

Constant	Value	Defined in
Lesion persistence threshold	14 days	<code>RiskCatalog.persistenceThresholdDays</code>
Lower-risk band maximum	4	<code>RiskCatalog.lowerRiskMaxScore</code>
Increased-risk band maximum	9	<code>RiskCatalog.increasedRiskMaxScore</code>
Maximum possible score	27	<code>RiskCatalog.maxPossibleScore</code>
Attendance window	14 days	<code>ClinicianAssessmentRecord.attendanceWindowDays</code>
Photograph size ceiling	900 KiB (Firestore limit 1 MiB)	<code>PhotoDocumentStore.maxBytes</code>
Photograph capture size	1024 × 1024, quality 55	<code>lesion_screen.dart</code>
Follow-up: professional check	7 days	<code>FollowUpPolicy</code>
Follow-up: observe and review	14 days from first noticed	<code>FollowUpPolicy</code>
Follow-up: higher risk	14 days	<code>FollowUpPolicy</code>
Follow-up: increased risk	30 days	<code>FollowUpPolicy</code>
Reminder lead time	2 days before, 09:00	<code>ReminderService</code>
Patient ID format	<code>0C-{year}-{6 alphanumeric}</code>	<code>AuthRepository</code>
Timezone	<code>Asia/Kolkata</code>	<code>ReminderService</code>